DISPROOF OF THE LOWER-DENSITY VARIANT OF ERDŐS PROBLEM 741

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For a set $S \subseteq \mathbb{N}$, write

$$\mathrm{lowerDensity}(S) := \liminf_{n\to\infty} \frac{|S \cap [0,n)|}{n}.$$

In connection with Erdős Problem 741 [1, 2], where Erdős asked whether $\overline{d}(A+A) > 0$ implies a decomposition $A = A_1 \sqcup A_2$ with $\overline{d}(A_1+A_1) > 0$ and $\overline{d}(A_2+A_2) > 0$, the corresponding lower-density question asks whether $\mathrm{lowerDensity}(A+A) > 0$ always allows a disjoint decomposition $A = A_1 \sqcup A_2$ with $\mathrm{lowerDensity}(A_1+A_1) > 0$ and $\mathrm{lowerDensity}(A_2+A_2) > 0$.

Theorem. *There exists $A \subseteq \mathbb{N}$ with* $\mathrm{lowerDensity}(A+A) > 0$ *such that for every decomposition $A = A_1 \cup A_2$ with $A_1 \cap A_2 = \emptyset$, either* $\mathrm{lowerDensity}(A_1+A_1) = 0$ *or* $\mathrm{lowerDensity}(A_2+A_2) = 0$.

Consequently, the lower-density variant of Erdős Problem 741 has a negative answer.

Correspondence with the Lean formalization. The proof below is the one formalized in `FormalConjectures/ErdosProblems/741.lean`, step by step, with the same parameters, sets, and inequalities in pure $\mathbb{N}$ arithmetic (passing through $\mathbb{R}$ only at the initial and final lowerDensity interface). All variables below are natural numbers unless noted otherwise. The correspondences with Lean names are:

This note	Lean
$\mathrm{lowerDensity}(S)$; $\lvert S \cap [0,n)\rvert$	`Set.lowerDensity S`; `(S ∩ Iio n).ncard`
Lemma 1 (a), (b)	`lowerDensity_pos_of_forall_ge`, `exists_eventually_ge_of_lowerDensity_pos`
$\mathrm{digit}(B,x,t)$; $\{x < B^d : \forall s < d,\ \mathrm{digit}(B,x,s) < K\}$	`digit B x t`; `boundedDigits B K d` (`boundedDigits_spec`, `boundedDigits_card`)
no-carry digit shift; $(x+a) \bmod m = 0$ criterion; $B^n \le (B-1)^n + nB^{n-1}$	`digit_add_mul_pow`, `digit_add_pow`; `add_mod_eq_zero_iff`; `pow_le_pred_pow_add`
q_k, m_k, $d_{0,k}$, d_k, B_k, Q_k	`qParam k`, `mParam k`, `d0Param k`, `dParam k`, `BParam k`, `QParam k`
configuration $\mathrm{cfg} = (\iota, \sigma)$; $R_k = \lvert\mathrm{Cfg}(k)\rvert$; cfg_r	`Cfg k`; `RParam k` (`RParam_pos`); `getCfg k r` (`exists_getCfg_eq`)
M_k, $L_{0,k}$, $D_{1/2,k}$, $D_{0,k}$, P_k, $N_{0,k}$, $N_{k,r}$	`M_end k`, `L0 k`, `DHalf k`, `D0 k`, `PParam k`, `N0 k`, `NScale k r` (via `stepScale`)
scale inequalities and divisibilities	`scale_chain`, `M_end_ge_16`, `NScale_strictMono`, `M_end_strictMono`, `PParam_dvd_NScale`, `QParam_dvd_D0`, `BParam_dvd_QParam`
$\mathrm{coordIdx}_k(j,u)$; $\mathrm{coeffPair}_k(\mathrm{cfg},(j,u))$; hiForm_k; $\mathrm{linForm}_k$	`coordIdx k j u`; `coeffPair k cfg p` (`coeffPair_apply`); `hiForm k cfg q`; `linForm k cfg x`
Q_k-periodicity and hub shift of $\mathrm{linForm}_k$	`linForm_add_mul_QParam`; `linForm_hub_add`
I_k, J_k, $\mathrm{hub}_k(j,u)$, H_k, $\mathrm{hubSet}_k(\mathrm{cfg})$, $W_{k,r}$, G_k, A	`Ik k`, `Jk k`, `hub k j u`, `Hk k`, `hubSet k cfg`, `Wk k r`, `GadgetK k`, `CounterexampleA`
$G_k \subseteq [M_k, M_{k+1})$; Lemma 2	`GadgetK_subset_Ico`; `exists_cfg_of_partition`
$\mathrm{mkW}_k(\mathrm{cfg},p)$; $\mathrm{baseWFinset}_k(\mathrm{cfg})$; Lemma 3	`mkW k cfg p` (`mkW_props`); `baseWFinset k cfg` (`mem_baseWFinset`); `baseWFinset_card_bound`
Lemma 4; Lemma 5; Lemma 6	`sumset_scale_bound`; `sumset_bridge_J_I_bound`; `lowerDensity_CounterexampleA_add_pos`
Lemma 7; Lemma 8; Theorem	`mem_target_lt_NScale_cases`; `target_sumset_ncard_bound`; `erdos_741.variants.lower`

Lemma 1. *Let $S \subseteq \mathbb{N}$.*

(a) *If $c \in \mathbb{R}$ satisfies $c > 0$ and there is $n_0 \in \mathbb{N}$ such that $cn \le |S \cap [0,n)|$ for all $n \ge n_0$, then $0 < \operatorname{lowerDensity}(S)$.*
(b) *If $0 < \operatorname{lowerDensity}(S)$, then there exist $K, N_0 \in \mathbb{N}$ such that $n \le K\,|S \cap [0,n)|$ for all $n \ge N_0$.*

Proof. Part (a) is immediate from the definition of $\liminf$, since $|S \cap [0,n)|/n \le 1$ is bounded above and eventually $\ge c$ for $n \ge \max(n_0, 1)$. For (b), pick $K \in \mathbb{N}$ with $1/\operatorname{lowerDensity}(S) < K$; then $1/K < \operatorname{lowerDensity}(S)$, so $|S \cap [0,n)|/n > 1/K$ for all sufficiently large $n \ge 1$. □

Base-B digits and bounded-digit fibers. For $B, x, t \in \mathbb{N}$, define $\operatorname{digit}(B,x,t) := \lfloor x/B^t \rfloor \bmod B$.

- If $B > 0$, $B^d \mid z$, and $t < d$, then $\operatorname{digit}(B, z+w, t) = \operatorname{digit}(B, w, t)$.
- If $B > 0$ and $\operatorname{digit}(B,w,t) + 1 < B$, then adding B^t to w produces no carry out of digit t: $\lfloor (w + B^t)/B^{t+1} \rfloor = \lfloor w/B^{t+1} \rfloor$, and for every $t' \in \mathbb{N}$,

$$\operatorname{digit}(B, w + B^t, t') = \operatorname{digit}(B, w, t') + \mathbf{1}_{t'=t}.$$

- For $K \le B$ and $d \in \mathbb{N}$, recursively define $\operatorname{BD}(B,K,0) := \{0\}$ and $\operatorname{BD}(B,K,d+1) := \{qB + r : q \in \operatorname{BD}(B,K,d),\ r < K\}$. By induction on d, every $x \in \operatorname{BD}(B,K,d)$ satisfies $x < B^d$ and $\operatorname{digit}(B,x,s) < K$ for all $s < d$, and $|\operatorname{BD}(B,K,d)| = K^d$.
- For $B \ge 1$ and $n \in \mathbb{N}$, induction on n gives $B^n \le (B-1)^n + nB^{n-1}$.

Parameters, configurations, scales, and linear forms. For each stage $k \in \mathbb{N}$, put

$$q_k := k+2, \quad m_k := 2q_k = 2k+4, \quad d_{0,k} := 2m_k^2, \quad d_k := d_{0,k} + 1, \quad B_k := 8m_k^3, \quad Q_k := B_k^{d_k}.$$

Let $\operatorname{Cfg}(k)$ be the finite type of pairs $\mathrm{cfg} = (\iota, \sigma)$ consisting of an injection $\iota : \{0, \dots, q_k - 1\} \hookrightarrow \{0, \dots, m_k - 1\}$ and, for each $i < q_k$, an injection $\sigma_i : \{0, \dots, m_k - 1\} \hookrightarrow \{0, \dots, 2m_k - 1\}$. Set $R_k := |\operatorname{Cfg}(k)| > 0$ and enumerate $\operatorname{Cfg}(k) = \{\mathrm{cfg}_0, \dots, \mathrm{cfg}_{R_k - 1}\}$.

Define the stage boundaries M_k recursively by $M_0 := 16$ and, given M_k,

$$L_{0,k} := 4M_kQ_k, \qquad D_{1/2,k} := 2m_kL_{0,k}, \qquad D_{0,k} := 2D_{1/2,k} = 4m_kL_{0,k},$$

$$P_k := 2m_kD_{0,k}, \qquad N_{0,k} := 10m_kP_k = 80m_k^3L_{0,k}, \qquad N_{k,r} := (4m_k)^r N_{0,k},$$

and $M_{k+1} := N_{k,R_k}$. Since $m_k \ge 4$ and $M_k \ge 16$, we have $B_k \mid Q_k \mid L_{0,k} \mid D_{1/2,k} \mid D_{0,k} \mid P_k \mid N_{k,r}$ and

$$4M_k \le L_{0,k},\ \ Q_k < L_{0,k},\ \ 2L_{0,k} < D_{1/2,k},\ \ (m_k+1)D_{0,k} < P_k,\ \ 40P_k \le N_{0,k} \le N_{k,r} < N_{k,r+1}.$$

In particular, $M_0 < M_1 < M_2 < \cdots$, so $k \le M_k$ for every k.

Fix the canonical bijection $\phi_k : \{0, \dots, m_k - 1\} \times \{0, \dots, 2m_k - 1\} \xrightarrow{\sim} \{0, \dots, d_{0,k} - 1\}$ (`finProdFinEquiv`), and put $\operatorname{coordIdx}_k(j,u) := \phi_k(j,u) + 1 \in \{1, \dots, d_{0,k}\}$. For $\mathrm{cfg} = (\iota, \sigma) \in \operatorname{Cfg}(k)$, define $\operatorname{coeffPair}_k(\mathrm{cfg}, (j,u)) := v$ if $(j,u) = (\iota(i), \sigma_i(v))$ for some (necessarily unique) (i,v), and 0 otherwise. Set

$$\operatorname{hiForm}_k(\mathrm{cfg}, q) := \sum_{j < m_k} \sum_{u < 2m_k} \operatorname{coeffPair}_k(\mathrm{cfg}, (j,u))\ \operatorname{digit}(B_k, q, \phi_k(j,u)),$$

$$\operatorname{linForm}_k(\mathrm{cfg}, x) := (x \bmod B_k) + \operatorname{hiForm}_k(\mathrm{cfg}, \lfloor x/B_k \rfloor).$$

Because $Q_k = B_k^{d_{0,k}} B_k$, if $Q_k \mid z$ then $(z+w) \bmod B_k = w \bmod B_k$ and, for every $s < d_{0,k}$, $\operatorname{digit}(B_k, \lfloor (z+w)/B_k \rfloor, s) = \operatorname{digit}(B_k, \lfloor w/B_k \rfloor, s)$, whence $\operatorname{linForm}_k(\mathrm{cfg}, z+w) = \operatorname{linForm}_k(\mathrm{cfg}, w)$.

The counterexample set A. For each $k \in \mathbb{N}$, define:

$$\begin{aligned}
I_k &:= [M_k, L_{0,k}),\\
J_k &:= \{qL_{0,k} : 1 \le q < 80m_k^3\},\\
\mathrm{hub}_k(j,u) &:= (j+1)D_{0,k} + B_k^{\mathrm{coordIdx}_k(j,u)}, \qquad H_k := \{\mathrm{hub}_k(j,u) : j < m_k,\ u < 2m_k\},\\
\mathrm{hubSet}_k(\mathrm{cfg}) &:= \{\mathrm{hub}_k(\iota(i), \sigma_i(v)) : i < q_k,\ v < m_k\},\\
W_{k,r} &:= \big\{w \in [N_{k,r}, N_{k,r+1}) : w \bmod P_k < D_{1/2,k},\ \mathrm{linForm}_k(\mathrm{cfg}_r, w) \bmod m_k = 0,\\
&\qquad\qquad \forall s < d_{0,k},\ \mathrm{digit}(B_k, \lfloor w/B_k \rfloor, s) < B_k - 1\big\} \qquad (r < R_k),\\
G_k &:= I_k \cup J_k \cup H_k \cup \bigcup_{r<R_k} W_{k,r}, \qquad A := [0,16) \cup \bigcup_{k\in\mathbb{N}} G_k.
\end{aligned}$$

Since $B_k^{\mathrm{coordIdx}_k(j,u)} \le B_k^{d_{0,k}} < Q_k < D_{0,k}$, every $\mathrm{hub}_k(j,u)$ satisfies $D_{0,k} \le \mathrm{hub}_k(j,u) < (m_k+1)D_{0,k} < P_k < N_{0,k}$, and hence $G_k \subseteq [M_k, M_{k+1})$. Moreover, if $\mathrm{digit}(B_k, \lfloor w/B_k \rfloor, s) < B_k - 1$ for all $s < d_{0,k}$, writing $\mathrm{hub}_k(j,u) + w = (j+1)D_{0,k} + (w + B_k^{\phi_k(j,u)} B_k)$ and using $Q_k \mid D_{0,k}$ together with the no-carry digit shift at $s_0 = \phi_k(j,u)$ gives

$$\mathrm{linForm}_k(\mathrm{cfg}, \mathrm{hub}_k(j,u) + w) = \mathrm{linForm}_k(\mathrm{cfg}, w) + \mathrm{coeffPair}_k(\mathrm{cfg}, (j,u)). \tag{1}$$

Lemma 2. *If $A = A_1 \cup A_2$, then for every $k \in \mathbb{N}$ there exists $r < R_k$ such that* $\mathrm{hubSet}_k(\mathrm{cfg}_r) \subseteq A_1$ *or* $\mathrm{hubSet}_k(\mathrm{cfg}_r) \subseteq A_2$.

Proof. For each $j < m_k$, the $2m_k$ elements $\{\mathrm{hub}_k(j,u) : u < 2m_k\} \subseteq H_k \subseteq A = A_1 \cup A_2$ satisfy $|\{u : \mathrm{hub}_k(j,u) \in A_1\}| \ge m_k$ or $|\{u : \mathrm{hub}_k(j,u) \in A_2\}| \ge m_k$. Since $m_k = 2q_k$, at least q_k indices $j < m_k$ satisfy the first alternative, or at least q_k satisfy the second. Taking the order-preserving embeddings of a q_k-element subset of blocks and of an m_k-element subset of slots in each chosen block yields $\mathrm{cfg} \in \mathrm{Cfg}(k)$ with $\mathrm{hubSet}_k(\mathrm{cfg}) \subseteq A_1$ or $\subseteq A_2$. □

Lemma 3. *For $k \in \mathbb{N}$, $\mathrm{cfg} \in \mathrm{Cfg}(k)$, and $p = (p_1, p_2) \in \mathbb{N}^2$, define*

$$\mathrm{mkW}_k(\mathrm{cfg}, p) := p_1 B_k + p_2 m_k + \big((m_k - \mathrm{hiForm}_k(\mathrm{cfg}, p_1) \bmod m_k) \bmod m_k\big),$$

and let $\mathrm{baseWFinset}_k(\mathrm{cfg})$ *be the image of* $\mathrm{BD}(B_k, B_k - 1, d_{0,k}) \times \{0, \dots, 8m_k^2 - 1\}$ *under* $\mathrm{mkW}_k(\mathrm{cfg}, \cdot)$. *Every $x \in \mathrm{baseWFinset}_k(\mathrm{cfg})$ satisfies $x < Q_k$,* $\mathrm{digit}(B_k, \lfloor x/B_k \rfloor, s) < B_k - 1$ *for all $s < d_{0,k}$, and* $\mathrm{linForm}_k(\mathrm{cfg}, x) \bmod m_k = 0$*; moreover,*

$$3Q_k \le 4m_k \,|\, \mathrm{baseWFinset}_k(\mathrm{cfg})|.$$

Proof. For $p_2 < 8m_k^2$, the low part $\mathrm{lo} := p_2 m_k + ((m_k - \mathrm{hiForm}_k(\mathrm{cfg}, p_1) \bmod m_k) \bmod m_k)$ is $< 8m_k^3 = B_k$, so $\lfloor \mathrm{mkW}_k(\mathrm{cfg}, p)/B_k \rfloor = p_1$ and $\mathrm{mkW}_k(\mathrm{cfg}, p) \bmod B_k = \mathrm{lo}$. This gives the three membership properties and shows $\mathrm{mkW}_k(\mathrm{cfg}, \cdot)$ is injective on the product, so $|\,\mathrm{baseWFinset}_k(\mathrm{cfg})| = (B_k - 1)^{d_{0,k}} \cdot 8m_k^2$. Since $4d_{0,k} B_k^{d_{0,k}-1} = 8m_k^2 B_k^{d_{0,k}-1} \le B_k^{d_{0,k}}$, the inequality $B_k^{d_{0,k}} \le (B_k - 1)^{d_{0,k}} + d_{0,k} B_k^{d_{0,k}-1}$ yields $3B_k^{d_{0,k}} \le 4(B_k - 1)^{d_{0,k}}$. Multiplying by $B_k = 8m_k^3$ gives $3Q_k \le 4m_k \,|\, \mathrm{baseWFinset}_k(\mathrm{cfg})|$. □

Lemma 4. *If $r < R_k$ and $4N_{k,r} \le n \le N_{k,r+1}$, then $3n \le 64\,|(A + A) \cap [0, n)|$.*

Proof. Let $\mathrm{cfg} = (\iota, \sigma) := \mathrm{cfg}_r$, $p_{\mathrm{start}} := N_{k,r}/P_k \ge 40$, $Q_n := \lfloor n/P_k \rfloor \ge 4p_{\mathrm{start}} \ge 160$, $\mathrm{numP} := Q_n - p_{\mathrm{start}} - 1$, and $\mathrm{numB} := 8m_k M_k$ (so $D_{1/2,k} = \mathrm{numB} \cdot Q_k$). For $(i,v) \in \{0, \dots, q_k - 1\} \times \{0, \dots, m_k - 1\}$, $t < \mathrm{numP}$, $b < \mathrm{numB}$, and $x \in \mathrm{baseWFinset}_k(\mathrm{cfg})$, put $w(t,b,x) := (p_{\mathrm{start}} + t)P_k + bQ_k + x$ and

$$\Phi(i, v, t, b, x) := \mathrm{hub}_k(\iota(i), \sigma_i(v)) + w(t, b, x).$$

Since $Q_k \mid P_k$, we have $w(t,b,x) \in W_{k,r}$, and $\Phi(i,v,t,b,x) < (p_{\mathrm{start}} + t + 2)P_k \le Q_n P_k \le n$, so Φ maps into $(A + A) \cap [0, n)$. Moreover, Φ is injective: writing $z = \Phi(i,v,t,b,x)$ and $y = bQ_k + x < D_{1/2,k}$, we have $B_k^{\mathrm{coordIdx}_k(\iota(i),\sigma_i(v))} + y < Q_k + D_{1/2,k} < D_{0,k}$ and $(\iota(i)+1)D_{0,k} + D_{0,k} \le (m_k + 1)D_{0,k} < P_k$, so $\lfloor z/P_k \rfloor = p_{\mathrm{start}} + t$ recovers t, and $\lfloor (z \bmod P_k)/D_{0,k} \rfloor = \iota(i) + 1$ recovers i by injectivity of ι. Next, (1) gives $\mathrm{linForm}_k(\mathrm{cfg}, z) \bmod m_k = v$, recovering v; subtracting

$\mathrm{hub}_k(\iota(i), \sigma_i(v)) + (p_{\mathrm{start}} + t)P_k$ recovers $bQ_k + x$, hence $b = \lfloor (bQ_k + x)/Q_k \rfloor$ and x. Thus the image F has size $q_k m_k \,\mathrm{numP}\,\mathrm{numB} \,|\, \mathrm{baseWFinset}_k(\mathrm{cfg})|$. Since $Q_n + 1 \le 2\,\mathrm{numP}$ and $P_k = 32m_k^3 M_k Q_k$, Lemma 3 gives

$$3n \le 3(Q_n + 1)P_k \le 6\,\mathrm{numP}\,P_k = 64m_k^2\,\mathrm{numP}\,M_k(3Q_k) \le 64\,|F| \le 64\,|(A + A) \cap [0, n)|. \quad \square$$

Lemma 5. *If $2L_{0,k} \le n \le N_{0,k}$, then $3n \le 64\,|(A + A) \cap [0, n)|$.*

Proof. Set $Q_L := \lfloor n/L_{0,k} \rfloor$; then $2 \le Q_L \le 80m_k^3$. For $1 \le q < Q_L$ and $x \in [M_k, L_{0,k})$, we have $qL_{0,k} \in J_k \subseteq A$, $x \in I_k \subseteq A$, and $qL_{0,k} + x < Q_L L_{0,k} \le n$. The map $(q, x) \mapsto qL_{0,k} + x$ is injective because $x < L_{0,k}$, producing $(Q_L - 1)(L_{0,k} - M_k)$ sums in $(A + A) \cap [0, n)$. Since $Q_L + 1 \le 3(Q_L - 1)$ and $3L_{0,k} \le 4(L_{0,k} - M_k)$, we obtain $3n \le 3(Q_L + 1)L_{0,k} \le 12(Q_L - 1)(L_{0,k} - M_k) \le 64\,|(A + A) \cap [0, n)|$. $\square$

Lemma 6. $\mathrm{lowerDensity}(A + A) > 0$.

Proof. By Lemma 1 (a) with $c = 3/256$ and $n_0 = 16$, it suffices to show $3n \le 256\,|(A+A)\cap[0, n)|$ for every $n \ge 16$. Choose k with $M_k \le n < M_{k+1}$. Note that $0 \in [0, 16) \subseteq A$.

- If $n < 4M_k$: for $k = 0$, $n < 64$ and $0 = 0 + 0 \in (A + A) \cap [0, n)$, so $3n < 192 < 256$. For $k = k' + 1$, $M_k = N_{k',R_{k'}-1+1} \ge 4N_{k',R_{k'}-1}$, so Lemma 4 at $(k', R_{k'} - 1)$ gives $3M_k \le 64\,|(A+A)\cap[0, M_k)|$, and monotonicity gives $3n < 12M_k \le 256\,|(A+A)\cap[0, n)|$.
- If $4M_k \le n \le 2L_{0,k}$: the sums $0+z$ for $z \in [M_k, \min(n, L_{0,k})) \subseteq I_k$ lie in $(A+A)\cap[0, n)$, giving $\min(n, L_{0,k}) - M_k$ elements; since $4M_k \le L_{0,k}$, we have $3n \le 256(\min(n, L_{0,k}) - M_k)$.
- If $2L_{0,k} < n \le N_{0,k}$: Lemma 5 gives $3n \le 64\,|(A + A) \cap [0, n)|$.
- If $N_{0,k} < n < M_{k+1} = N_{k,R_k}$: pick $r < R_k$ with $N_{k,r} \le n < N_{k,r+1}$. If $4N_{k,r} \le n$, Lemma 4 gives $3n \le 64\,|(A+A)\cap[0, n)|$. If $n < 4N_{k,r}$, then $3N_{k,r} \le 64\,|(A+A)\cap[0, N_{k,r})|$ holds by Lemma 5 (when $r = 0$) or Lemma 4 at $r - 1$ (when $r \ge 1$), so monotonicity yields $3n < 12N_{k,r} \le 256\,|(A + A) \cap [0, n)|$. $\square$

Lemma 7. *Suppose $A_t \subseteq A$, $A_t \cap A_o = \emptyset$, $r < R_k$, and $\mathrm{hubSet}_k(\mathrm{cfg}_r) \subseteq A_o$. Every $x \in A_t$ with $x < N_{k,r+1}$ satisfies either $x \in W_{k,r}$, or $x < N_{k,r}$ together with one of:*

(i) $x \bmod P_k < D_{1/2,k}$, *(ii)* $x \in J_k$, *or (iii)* $x \in H_k \setminus \mathrm{hubSet}_k(\mathrm{cfg}_r)$.

Proof. Since $G_{k'} \subseteq [M_{k'}, M_{k'+1})$ and $N_{k,r+1} \le M_{k+1}$, no element of $G_{k'}$ with $k' > k$ is $< N_{k,r+1}$. Elements of $[0, 16)$, of $G_{k'}$ with $k' < k$, and of I_k are all $< L_{0,k} < D_{1/2,k} < P_k < N_{k,r}$, giving (i). On stage k, elements of J_k satisfy $x < N_{0,k} \le N_{k,r}$, giving (ii); elements of $H_k \cap A_t$ satisfy $x < P_k < N_{k,r}$ and cannot lie in $\mathrm{hubSet}_k(\mathrm{cfg}_r) \subseteq A_o$ since $A_t \cap A_o = \emptyset$, giving (iii); finally, for $x \in W_{k,u}$, $u > r$ is impossible as $x < N_{k,r+1} \le N_{k,u}$, $u = r$ gives $x \in W_{k,r}$, and $u < r$ gives $x < N_{k,u+1} \le N_{k,r}$ and $x \bmod P_k < D_{1/2,k}$, which is (i). $\square$

Lemma 8. *Under the hypotheses of Lemma 7, let $N := N_{k,r+1}$. Then*

$$m_k\,|(A_t + A_t) \cap [0, N)| \le 2N.$$

Proof. Put $\mathrm{cfg} := \mathrm{cfg}_r$. Define three finite sets:

$$E_1 := [0, 2N_{k,r}),$$
$$E_2 := \{p_1 P_k + p_2 : p_1 < N/P_k,\ p_2 < D_{0,k}\},$$
$$E_3 := \{\mathrm{mkW}_k(\mathrm{cfg}, (q, u)) : q < N/B_k,\ u < 8m_k^2\}.$$

Let $z = x + y < N$ with $x, y \in A_t$, and apply Lemma 7 to x and y.

- If $x < N_{k,r}$ and $y < N_{k,r}$, then $z < 2N_{k,r}$, so $z \in E_1$.
- If $x \bmod P_k < D_{1/2,k}$ and $y \bmod P_k < D_{1/2,k}$ (covering $x, y \in W_{k,r}$ as well as case (i) paired with $W_{k,r}$), then $z \bmod P_k < 2D_{1/2,k} = D_{0,k}$, and $P_k \mid N$ implies $\lfloor z/P_k \rfloor < N/P_k$, so $z \in E_2$.
- In the remaining mixed cases, one summand (say y) lies in $W_{k,r}$ and the other (x) lies in J_k or $H_k \setminus \mathrm{hubSet}_k(\mathrm{cfg})$. If $x = qL_{0,k} \in J_k$, then $Q_k \mid x$, so $\mathrm{linForm}_k(\mathrm{cfg}, x + y) = \mathrm{linForm}_k(\mathrm{cfg}, y)$. If $x = \mathrm{hub}_k(j, u) \in H_k \setminus \mathrm{hubSet}_k(\mathrm{cfg})$, then (j, u) is not in the image of

cfg, so $\mathrm{coeffPair}_k(\mathrm{cfg},(j,u))=0$, and (1) gives $\mathrm{linForm}_k(\mathrm{cfg},x+y)=\mathrm{linForm}_k(\mathrm{cfg},y)$. In both cases $\mathrm{linForm}_k(\mathrm{cfg},z) \bmod m_k=0$. Writing $q=\lfloor z/B_k\rfloor < N/B_k$ and $\mathrm{lo}=z \bmod B_k < B_k=8m_k^3$, the congruence $(\mathrm{lo}+\mathrm{hiForm}_k(\mathrm{cfg},q)) \bmod m_k=0$ forces $\mathrm{lo} \bmod m_k = (m_k-\mathrm{hiForm}_k(\mathrm{cfg},q) \bmod m_k) \bmod m_k$, so $z=\mathrm{mkW}_k(\mathrm{cfg},(q,\lfloor \mathrm{lo}/m_k\rfloor))\in E_3$.

Hence $(A_t+A_t)\cap[0,N)\subseteq E_1\cup E_2\cup E_3$. Since $2m_k|E_1|=4m_kN_{k,r}=N$, $2m_k|E_2|\le 2m_k(N/P_k)D_{0,k}=N$, and $m_k|E_3|\le m_k(N/B_k)(8m_k^2)=N$, we obtain $2m_k\,|(A_t+A_t)\cap[0,N)|\le 2m_k(|E_1|+|E_2|+|E_3|)\le 4N$, so $m_k\,|(A_t+A_t)\cap[0,N)|\le 2N$. □

Proof of the Theorem. By Lemma 6, $\mathrm{lowerDensity}(A+A)>0$. Suppose for contradiction that $A=A_1\cup A_2$ with $A_1\cap A_2=\emptyset$, $\mathrm{lowerDensity}(A_1+A_1)>0$, and $\mathrm{lowerDensity}(A_2+A_2)>0$. Lemma 1 (b) gives (K_1,N_1) and (K_2,N_2) with $n\le K_i\,|(A_i+A_i)\cap[0,n)|$ for all $n\ge N_i$ $(i=1,2)$. Set

$$k:=\max\big(\max(N_1,N_2),\,\max(K_1,K_2)\big).$$

By Lemma 2, there is $r<R_k$ such that $\mathrm{hubSet}_k(\mathrm{cfg}_r)\subseteq A_1$ or $\mathrm{hubSet}_k(\mathrm{cfg}_r)\subseteq A_2$. Put $(A_t,A_o,K_t,N_t):=(A_2,A_1,K_2,N_2)$ in the first case and (A_1,A_2,K_1,N_1) in the second, and set $N:=N_{k,r+1}$ and $c:=|(A_t+A_t)\cap[0,N)|$. Since $N_t\le k\le M_k<N_{0,k}\le N$, we have $N\le K_tc$ (which with $N>0$ forces $c>0$), while Lemma 8 gives $m_kc\le 2N$. Because $K_t\le k$ and $m_k=2(k+2)$, we have $2K_t+1\le m_k$, so $(2K_t+1)c\le m_kc\le 2N\le 2K_tc$, contradicting $c>0$. □

References

[1] P. Erdős, *Some problems in number theory, combinatorics and combinatorial geometry*, Math. Pannon. **5** (1994), 261–269.

[2] *Erdős Problem #741*, `https://www.erdosproblems.com/741`.